

ARI 410/510 Lab 1: First ML Experiments (v1.1)

University of Michigan–Flint

Fall 2026

(See the course Canvas page for the due date. Feel free to ask questions on Discord in the `#homework-help` channel.)

1 Overview

In this lab-style assignment, you will compare several machine learning classifiers using the scikit-learn Python library. By now we have seen three classifiers in lecture: ***k*-nearest neighbors**, **logistic regression**, and **support vector machines**. This is your chance to get hands-on, train them on real data, and compare how they behave. This is a gentle introduction to using ML libraries and to the train / tune / test workflow. If you already have ML experience this may feel simple; if it is your first time, plan for some time to get comfortable.

A starter notebook is provided (see Canvas) that loads the data, makes a train/dev/test split, and runs one *k*-NN model. You are welcome to copy it and build from there and you are not expected to write everything from scratch. The point of the lab is to *try things and interpret them*, not to reproduce boilerplate.

Note on GenAI before you start: if you use any generative AI on this assignment, you will be asked to fill out a short structured statement about how you used it and are encouraged to *attach your chat logs*. Set up whatever you need to export those logs *before* you begin, so you are not trying to reconstruct them at the end. Full details are in Section 6.

2 The Data

We will work with this heart disease classification dataset from Kaggle. The goal is to predict whether a patient has heart disease from a set of clinical features.

A note on real data: this dataset is not perfectly clean. You will need to make some decisions. For example, how to handle missing values, and how to turn the target column into the label vector you want to predict (e.g., presence vs. absence of disease). How you handle these is part of your approach; document what you did. (The starter notebook makes one reasonable set of choices; you are free to change them.)

Because this is a medical diagnosis task, it is worth thinking about which kind of mistake is worse: a false positive or a false negative, and letting that inform how you judge your models later.

3 Instructions (shared)

1. Download the dataset as a CSV from the course Canvas page.
2. Load the data in Python, storing it as a matrix of features X and a vector of labels y . Clean and encode as needed (see above).
3. Split the data into **train**, **development** (dev), and **test** sets. You will train on the train set, compare configurations on the dev set, and touch the test set only once, at the very end. (Keeping the test set untouched while you tune is how you avoid fooling yourself: if you choose settings using the test set, your final number no longer estimates performance on new, unseen data.)

4. Track **precision**, **recall**, **F1-score**, and **accuracy** throughout.

The next part (how you set up and run the comparison) differs by course level. Note that students in ARI 410 can choose either path (and should note this in their report), students in 510 must follow the 510 path.

ARI 410 (Undergraduate): run and interpret

Compare the three classifiers from lecture: ***k*-NN**, **logistic regression**, and **SVM** using the hyperparameter settings below. For each model, run all five settings, training on the train set and evaluating on the dev set each time.

- ***k*-NN** (scale your features first): $k = 1, 3, 5, 11, 25$.
- **Logistic regression** (L2 penalty): $C = 0.01, 0.1, 1, 10, 100$.
- **SVM**: (linear, $C=1$), (linear, $C=10$), (RBF, $C=1$), (RBF, $C=10$), (RBF, $C=100$).

That is $3 \times 5 = 15$ runs. Record the best setting for each model, then take those three best configurations and evaluate each *once* on the test set. You decide what “best” means and document your reasoning.

ARI 510 (Graduate): design and defend

Choose **at least five classifiers** of your choice from sklearn’s supervised learning methods (you are not required to use *k*-NN, logistic regression, or SVM, though you may). Choose and justify a protocol:

- **State a hypothesis before you run anything.** Which model(s) do you expect to do best on this data, and why? (Reason about the feature space, linear vs. nonlinear separability, dataset size, and so on.)
- **Design the hyperparameter search yourself.** Which hyperparameters, which ranges, and why. Explore enough to make the comparison fair, but don’t feel that you need to do so many experiments that your run takes days to complete.
- **Justify your methodology:** how you split the data, and which metric you treat as decisive and why (considering the nature of the task).

Compare to simple baselines (410 and 510 students):

In addition to comparing multiple models, you should also include at least two simple baseline models (this is something you should get in the habit of always doing). The first should always predict the *majority* class present in the training data. The second should be a *random* guessing baseline. This can either be based on a theoretical value (e.g., the random guessing accuracy when predicting a fair die roll would be $1/6$) or based on a simulation (e.g., computationally produce 10,000 sets of random predictions for the test set and compute the average accuracy, or whichever metric you are targeting, achieved by these predictions).

After your comparison (410 and 510 students): the value of more data

Once you have identified your single best configuration, run one more short study. Retrain that best model on **25%**, **50%**, and **100%** of your training data (keeping the dev and test sets fixed), and **plot** its performance against the training-set size. The sample notebook will help you get started, but you should also add results on the test set and change the model to the one that worked the best for you. Then comment: how much of your result came from the model and its settings, and how much came simply from how much data the model got to see? You should end up with one small plot and a few sentences. (*510: you are encouraged to go further, e.g., a finer-grained learning curve, or an additional type of ablation, i.e., removal of some part of the overall approach to understand how much it helped, of your choosing.*)

4 Deliverables

Submit:

1. your written report (PDF);
2. the code you used (a `.py` or `.ipynb` file, or a link to a GitHub repository);
3. your generative AI statement and documentation (see Section 6).

5 Report Requirements

Most of your grade is the report’s reasoning, the experimental choices and what you make of the results, not the code. Code is checked for being readable and for doing what you say it does.

Everyone

1. **Introduction:** the task and data, and which classifiers you used.
2. **Approach:** how you set up and ran your comparison, including how you cleaned and encoded the data.
3. **Results:** present your dev-set results across configurations (a figure or a table), your final test-set results, and your data-value ablation plot. *Please do not paste screenshots from a terminal or IDE, format these as actual tables or figures.*
4. **Discussion:** Address, at a minimum, all of the following: what did “best” mean in your experiments, and why; which model worked best; what the ablation showed about the value of data; what you learned; what challenges you hit and how you handled them.

ARI 410 (Undergraduate), additionally

1. **Connecting to concepts:** explain the behavior you observed in terms of what we covered in lecture. Why might the RBF kernel have beaten the linear one (or not)? Did the choice of k matter, and why? How did logistic regression and SVM differ in what they did, and does that line up with how they are built?

ARI 510 (Graduate), additionally

Because your task was to *design* the study, your report accounts for that design rather than reporting a fixed run:

1. **Hypothesis:** stated up front (before Results); in the Discussion, say whether the results bore it out and what that tells you.
2. **Methodology justification:** defend your overall strategy and your choice of decisive metric.
3. **Error analysis:** examine where your best model fails, which cases, and whether there is a pattern, and what that suggests.

6 Generative AI Use Statement (required)

Every submission must include **one** of the following.

Option A: No AI. If you used no generative AI, state exactly: “*No generative AI was used during this assignment.*” You are done with this section. Please note that typing a query into Google and getting the automatic Gemini response (or something similar to that, which I will call “unintentional AI use”) doesn’t

need to be documented here. Only include cases where you actively used a GenAI tool to help in some way (including autocomplete for coding, ChatGPT chat interface, agentic assistant like Claude Code, etc.).

Option B: Structured statement. If you used any generative AI tool you deliberately turned to (ChatGPT, Claude, Copilot, Gemini as a chatbot, etc.), complete the table below with one row per phase of your work. The point is not to confess to using AI (it is allowed) but to show *where the thinking was*: research on knowledge work finds that when people use AI, the real effort shifts from producing an artifact to verifying, integrating, and deciding whether to trust the output.¹

For each phase, note: (1) whether you used AI, and if the answer is “yes”, fill in: (2) *how* which mode(s) (see below), and what you actually asked (feel free to make this cell of the table large to explain) and (3) what you did to check, correct, or override the output. The three “modes”:

- **Creation:** had the AI generate something from scratch (code, text, an idea).
- **Information:** I used it to explain, find, or summarize (e.g., “what does the SVM C parameter do?”).
- **Advice / feedback:** I used it to improve, check, or critique something you already had (e.g., “review this function,” “is F1 the right metric here?”).

Phase	Used AI?	How (mode + what you asked)	How you checked / corrected / overrode it
Planning & understanding			
Coding and Debugging			
Analysis & interpretation			
Report writing, formatting, editing			

[Optional, but encouraged] **Attach your logs.** For any deliberate chatbot or AI agent use, export and attach your chat transcript or coding session, or include a shared link. Set this up *before* you start (this is why it’s flagged in the Overview). If you used an inline tool that cannot export a clean log (e.g., editor autocomplete), just describe how you used it instead.

7 Other questions?

Please ask on Discord in the #homework-help channel.

¹Structure adapted from H.-P. Lee, A. Sarkar, L. Tankelevitch, I. Drosos, S. Rintel, R. Banks, & N. Wilson, “The Impact of Generative AI on Critical Thinking,” CHI 2025.